Question 1 : Create a Heatmap to analyze Treatment Cost by Department and Region.

A Heatmap is a visual representation of data that uses different colors to show high and low values. It helps identify which departments and regions have the highest treatment costs.

Steps to create a Heatmap in Power BI

1. Load the dataset containing the `Department`, `Region`, and `Treatment Cost` columns.
2. Open the Visualizations pane and select a Matrix visual.
3. Drag the fields into the following areas:
- Rows: Department
- Columns: Region
- Values: Treatment Cost (set aggregation to Sum)
4. Select the Matrix visual and open the Format visual pane.
5. Expand Cell elements or Conditional formatting, depending on your Power BI version.
6. Enable Background color for the Treatment Cost values.
7. Choose a color scale, for example:
    - Green — Low treatment cost
    - Yellow — Medium treatment cost
    - Red — High treatment cost

Example of a Heatmap

Treatment Cost by Department and Region

Illustrative data only — replace with your actual dataset values.

Department	North	South	East	West
Cardiology	85k	62k	95k	70k
Orthopedics	45k	78k	55k	90k
Neurology	72k	50k	88k	60k

Oncology	98k	85k	75k	92k

Low

Medium

High

DAX measure (optional)

If you need a measure to calculate the total treatment cost, use:

- Total Treatment Cost =
SUM('Healthcare'[Treatment Cost])

Replace `'Healthcare'` with your actual table name.

Interpretation

- Red cells indicate departments and regions with higher treatment costs.
- Green cells indicate lower treatment costs.
- The heatmap helps management identify expensive departments, compare regional spending, and investigate opportunities to control healthcare costs.

Conclusion: A heatmap makes it easy to compare treatment costs across departments and regions and quickly identify areas requiring further analysis.

Question 2 : Create a Highlight Table showing Satisfaction Score by Department.

A Highlight Table uses colors to emphasize high and low satisfaction scores across departments. In Power BI, you can create one using a Matrix visual with conditional formatting.

Steps to create the Highlight Table in Power BI

1. Load your healthcare dataset containing `Department` and `Satisfaction Score`.
2. In the Visualizations pane, select the Matrix visual.
3. Drag the fields into these areas:
    - Rows: Department
    - Values: Satisfaction Score
4. Set the aggregation for Satisfaction Score to Average if there are multiple records per department.
5. Select the Matrix visual and open Format visual.
6. Find Cell elements → Background color or the conditional formatting option for the Satisfaction Score field.

7. Enable background color and apply a color scale:
    - Green = High satisfaction
    - Yellow = Moderate satisfaction
    - Red = Low satisfaction

Example Highlight Table

Average Satisfaction Score by Department

Illustrative values on a 1–10 scale.

Department	Average Satisfaction Score	Rating
Cardiology	8.7	High
Orthopedics	7.2	Moderate
Neurology	6.1	Low
Pediatrics	9.1	High
Oncology	7.8	Moderate

The example uses illustrative scores; actual results depend on your dataset.

DAX Measure (Optional)

To calculate the average satisfaction score, create this measure:

- Average Satisfaction Score = AVERAGE('Healthcare'[Satisfaction Score])

Replace `'Healthcare'` with your actual table name.

Interpretation

- Green: Departments with higher patient satisfaction.

- Yellow: Departments with moderate satisfaction.
- Red: Departments with lower satisfaction that may need improvement.

Conclusion: The highlight table helps management compare patient satisfaction across departments, identify areas performing well, and prioritize improvements in departments with lower scores.

Is this conversation helpful so far?

Question 3 : Create a Bump Chart to show ranking of Departments based on Treatment Cost over months.

A Bump Chart displays how the rankings of different departments change over time based on their treatment costs. It helps identify which departments consistently have the highest costs and how their positions change from month to month.

Steps to create a Bump Chart in Power BI

1. Load your healthcare dataset containing `Department`, `Treatment Cost`, and `Date`.
2. Extract the month and year from the Date column.
3. Create a measure to calculate the total treatment cost:

- Total Treatment Cost =
SUM('Healthcare'[Treatment Cost])

4. Create a ranking measure using DAX:

```
- Department Rank =
- RANKX(
-     ALL('Healthcare'[Department]),
-     [Total Treatment Cost],
-     ,
-     DESC,
-     DENSE
)
```

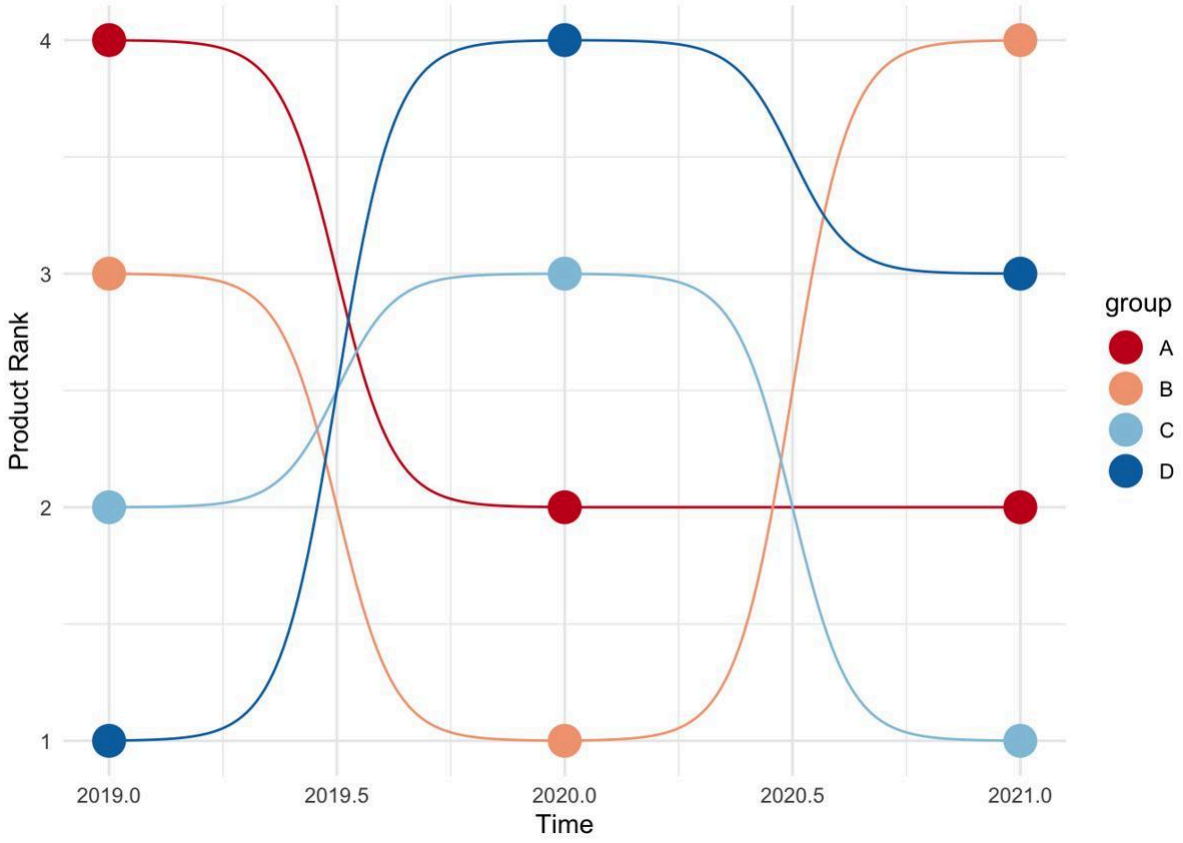
The department with the highest treatment cost receives rank 1.

5. In the Visualizations pane, select a Bump Chart custom visual from AppSource, if available, or use a line chart configured to display ranks.
6. Assign the fields:
    - X-axis: Month and Year
    - Y-axis: Department Rank
    - Legend: Department
    - Values: Department Rank
7. If using a line chart, reverse the Y-axis order so rank 1 appears at the top. Ensure the ranking measure responds to the selected month.

Example Bump Chart

Department Ranking by Treatment Cost

Illustrative rankings only; rank 1 means the highest treatment cost.



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Use your actual monthly data to produce the final chart.

Interpretation

- Rank 1: Department with the highest treatment cost in that month.
- Upward or downward movement: Indicates a change in department ranking, depending on the chart's axis orientation.
- Stable ranking: Suggests the department maintains a consistent position over time.
- Frequent ranking changes: May indicate changing treatment demand or monthly cost fluctuations.

Conclusion: The bump chart helps compare monthly treatment-cost rankings across departments, identify departments with consistently high costs, and investigate changes in healthcare spending over time.

Question 4 : Create a Butterfly Chart comparing Male vs Female patient count by Department.

A Butterfly Chart (also called a Tornado Chart) compares two groups side by side. In this case, it shows the number of male and female patients in each department, making it easy to compare patient distribution.

Steps to create a Butterfly Chart in Power BI

1. Load your healthcare dataset containing `Department`, `Gender`, and `Patient ID`.
2. Create a measure to count male patients:

```
- Male Patients =
- CALCULATE(
-     DISTINCTCOUNT('Healthcare'[Patient ID]),
-     'Healthcare'[Gender] = "Male"
)
```

3. Create a measure to count female patients:

```
- Female Patients =
- CALCULATE(
-     DISTINCTCOUNT('Healthcare'[Patient ID]),
-     'Healthcare'[Gender] = "Female"
)
```

4. In Power BI, open Visualizations → Get more visuals and search for a Butterfly Chart or Tornado Chart in AppSource.
5. Add the custom visual to your report and assign:
    - Category: Department
    - Male values: Male Patients
    - Female values: Female Patients
6. Configure the chart so male patient counts appear on the left and female patient counts on the right.
7. Add a title, data labels, and contrasting colors.

Example Butterfly Chart

Patient Count by Department and Gender

Illustrative data only.

Cardiology

70

55

Orthopedics

60

45

Neurology

40

50

Pediatrics

35

65

Oncology

45

40

Male patients (left)

Female patients (right)

Interpretation

- Cardiology and Orthopedics: More male than female patients in this example.
- Neurology and Pediatrics: More female than male patients.
- Oncology: Male and female patient counts are relatively similar.

Conclusion

The Butterfly Chart helps compare male and female patient counts across departments, identify differences in patient distribution, and support healthcare resource planning.

Note: The chart values above are illustrative. Use the actual dataset for accurate analysis.

Question 5 : Create a Dual Axis Chart to compare Average Age and Treatment Cost by Department.

A Dual Axis Chart in Power BI displays two different measures on the same chart using separate axes. It helps compare the average patient age with the treatment cost for each department.

Steps to create the chart in Power BI

1. Create the DAX measures

Average Age:

- Average Age = AVERAGE('Healthcare'[Age])

Total Treatment Cost:

- Total Treatment Cost = SUM('Healthcare'[Treatment Cost])

Replace `'Healthcare'` with your actual table name.

2. Insert a Line and Clustered Column Chart

In the Visualizations pane, select Line and clustered column chart and assign:

- X-axis: Department
- Column Y-axis: Total Treatment Cost
- Line Y-axis: Average Age

3. Format the chart

- Enable the secondary Y-axis for the line if available.
- Set the column axis to Treatment Cost.
- Set the line axis to Average Age.
- Use contrasting colors, such as blue columns for cost and an orange line for average age.
- Turn on data labels and add the title Average Age vs. Treatment Cost by Department.

Example chart

Average Age vs. Treatment Cost

Illustrative data only — not actual healthcare dataset results.

₹0₹250K₹500K₹750K₹1MCardiologyOrthopedicsNeurologyPediatricsOncology

Treatment cost shown as columns; average age shown separately below for illustration.

Average age by department (years)

0 years20 years40 years60 years80 yearsCardiologyOrthopedicsNeurologyPediatricsOncology

These separate charts illustrate the two measures. In Power BI, combine them into one dual-axis visual using the steps above.

Interpretation

- Departments with higher treatment costs may require greater financial resources.
- Average age can help identify departments treating older or younger patient populations.
- Comparing both measures can help investigate whether treatment costs vary with patient age.
- A high average age does not necessarily mean higher treatment costs; treatment type and patient complexity may also influence costs.

Conclusion

The Dual Axis Chart allows healthcare managers to compare average patient age and total treatment cost by department in one visual, helping them understand patient demographics and spending patterns.

Question 6 : Create a Sunburst Chart showing Department → Diagnosis hierarchy by Treatment Cost

A Sunburst Chart is a hierarchical visualization that displays data in concentric rings. The inner ring represents the Department, and the outer ring represents the Diagnosis within each department. The size of each segment indicates the treatment cost.

Steps to create the Sunburst Chart in Power BI

1. Load your healthcare dataset containing `Department`, `Diagnosis`, and `Treatment Cost`.
2. Open Visualizations → Get more visuals and search for Sunburst Chart in Microsoft AppSource.
3. Add the Sunburst visual to your report.
4. Drag the fields into the visual's field wells:

- Group / Category: Department
- Details / Child Category: Diagnosis
- Values: Treatment Cost (aggregation: Sum)

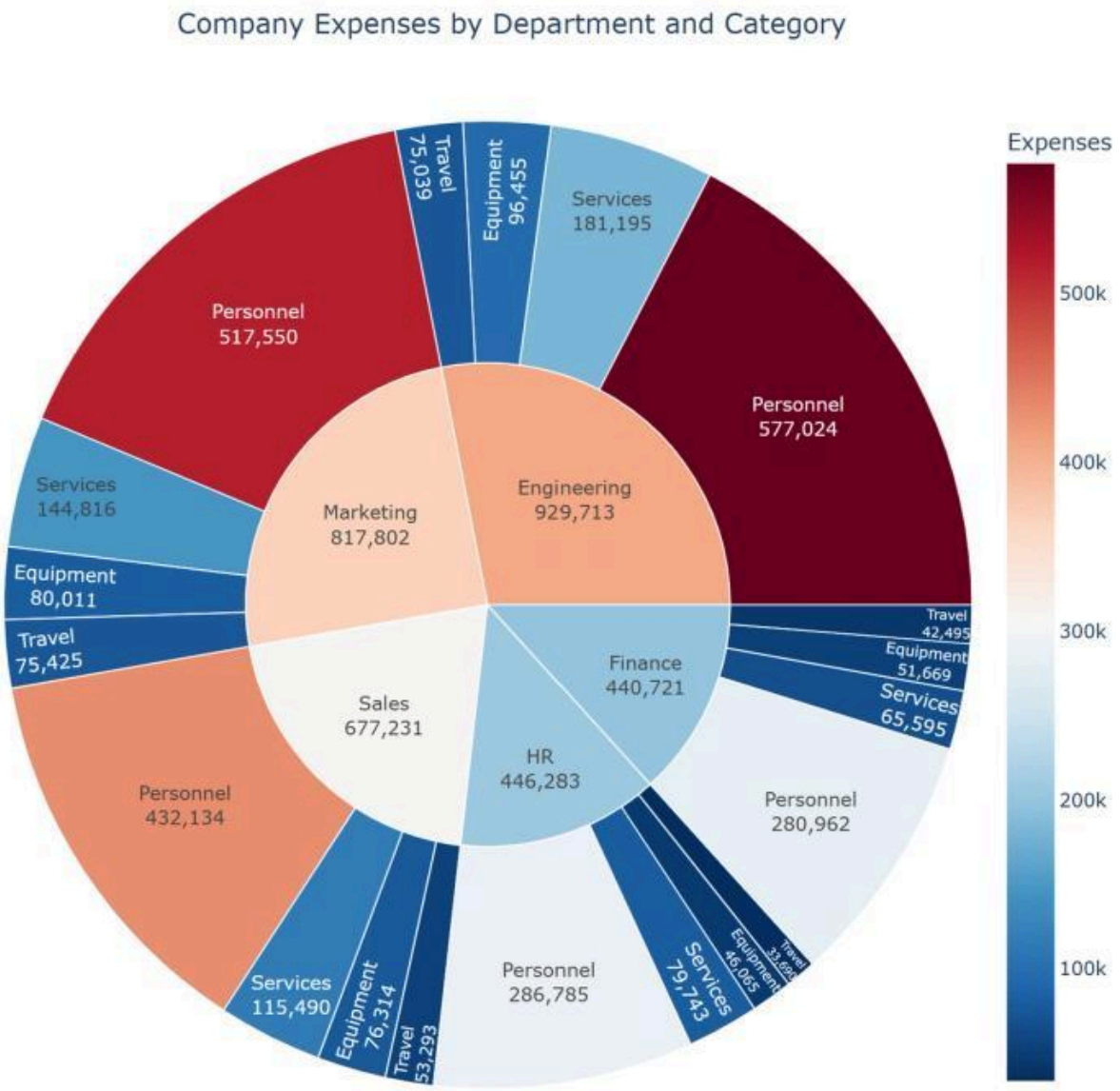
5. Format the chart:

- Set the inner ring to Department.
- Set the outer ring to Diagnosis.
- Enable data labels and tooltips.
- Use distinct colors for departments and their diagnoses.

Example Sunburst Chart

Treatment Cost by Department and Diagnosis

Illustrative visualization — actual values depend on your dataset.



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The inner ring shows departments; the outer ring breaks each department into diagnoses.

DAX Measure (Optional)

Create a measure to calculate the total treatment cost:

- Total Treatment Cost =
SUM('Healthcare'[Treatment Cost])

Replace `'Healthcare'` with your actual table name.

Interpretation

- Larger segments indicate higher treatment costs.
- Inner-ring segments allow you to compare costs across departments.
- Outer-ring segments help identify diagnoses contributing most to each department's cost.
- Departments with several large diagnosis segments may require further cost analysis.

Conclusion

The Sunburst Chart helps healthcare managers explore treatment costs at two levels—Department → Diagnosis—and identify the departments and diagnoses contributing most to total healthcare expenditure.

Question 7 : Create a Dual Axis Chart comparing Treatment Cost and Satisfaction Score by Department.

A Dual Axis Chart in Power BI compares two different measures using separate Y-axes. Here, it helps analyze the relationship between treatment costs and patient satisfaction across departments.

1. Create DAX measures

Total Treatment Cost

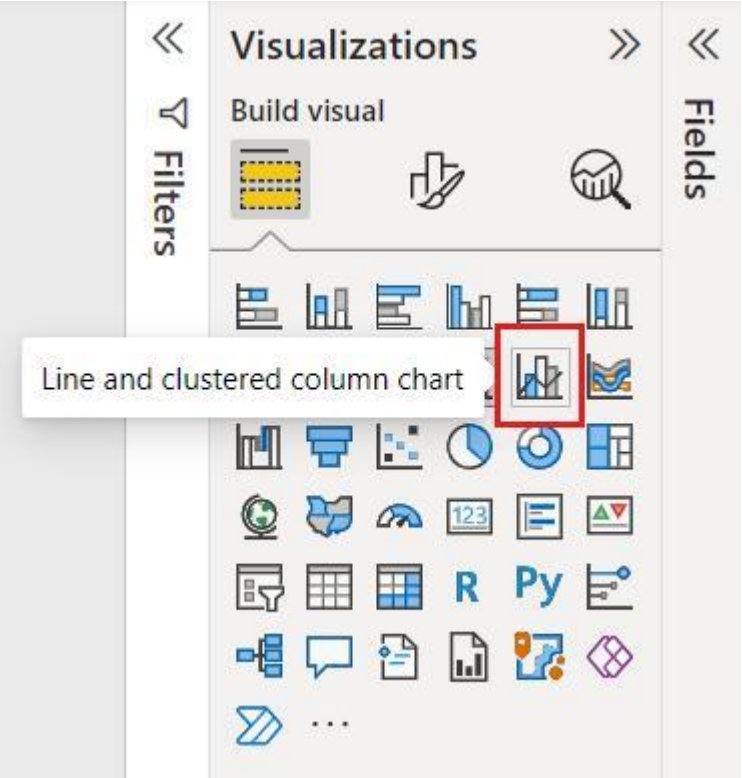
- Total Treatment Cost =
SUM('Healthcare'[Treatment Cost])

Average Satisfaction Score

- Average Satisfaction Score =
AVERAGE('Healthcare'[Satisfaction Score])

Replace `'Healthcare'` with your actual table name.

2. Create the chart in Power BI



Select the visual

In the Visualizations pane, choose Line and clustered column chart.

Assign the fields

- X-axis: Department
- Column Y-axis: Total Treatment Cost
- Line Y-axis: Average Satisfaction Score

Format the visual

1. Enable the secondary Y-axis for the line if available.
2. Use blue columns for treatment cost and an orange line for satisfaction score.
3. Enable data labels, tooltips, and a chart title.

3. Example chart

Treatment Cost vs. Satisfaction Score

Illustrative data only; values are not from your actual dataset.

₹0₹250K₹500K₹750K₹1MCardiologyOrthopedicsNeurologyPediatricsOncology

Total treatment cost by department (₹).

Average satisfaction score by department

02.557.510CardiologyOrthopedicsNeurologyPediatricsOncology

Satisfaction score on an illustrative 1–10 scale.

These separate charts illustrate the measures. In Power BI, combine them in one dual-axis visual using the field assignments above.

4. Interpretation

- High treatment cost and high satisfaction: May indicate that patients are receiving costly care with favorable satisfaction ratings.
- High treatment cost and low satisfaction: May indicate an area where service quality or cost-effectiveness needs investigation.
- Low treatment cost and high satisfaction: May suggest efficient service delivery.
- Low treatment cost and low satisfaction: May indicate opportunities to improve patient experience.

Conclusion

The Dual Axis Chart helps healthcare managers compare departmental treatment expenditure with patient satisfaction, identify potential improvement areas, and make more informed decisions about healthcare spending and service quality.

Question 8 : Create a Dashboard combining: Heatmap Dual Axis Chart Sunburst Chart Butterfly Chart

1. Dashboard layout

Here is the recommended layout for your Power BI report:

Healthcare Analytics Dashboard

Treatment Cost • Patient Satisfaction • Diagnosis • Gender

1. Treatment Cost Heatmap

Department × Region

Cardiology

Orthopedics

Neurology

Colors highlight low-to-high costs.

2. Dual Axis Chart

Treatment Cost and Satisfaction

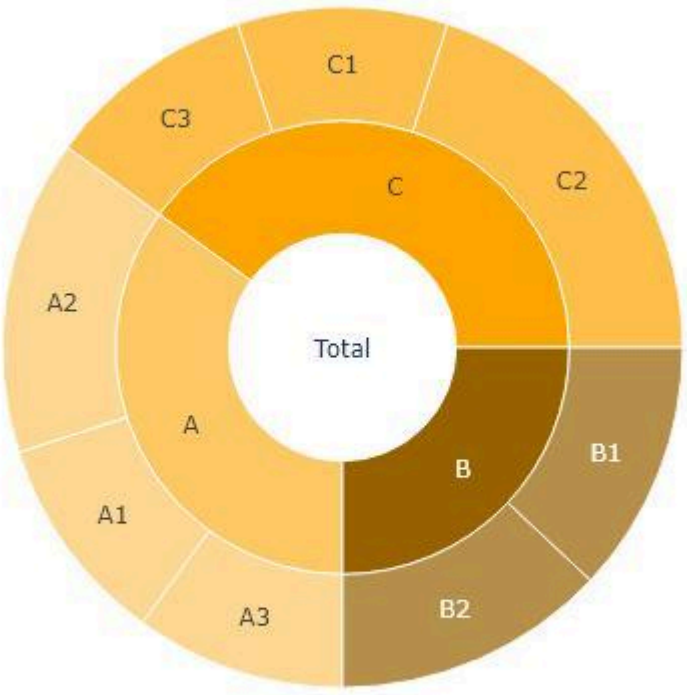
₹0₹250K₹500K₹750K₹1MCardioOrthoNeuroPediatrics

Illustrative treatment costs; satisfaction is shown separately below.

Combine columns and a satisfaction line in Power BI.

3. Sunburst Chart

Department → Diagnosis



Segment size represents treatment cost.

4. Butterfly Chart

Male vs. Female Patients

Cardio

Ortho

Neuro

■ Male

■ Female

Mockup only: the values and visual styling are illustrative, not connected to your dataset.

2. Steps to build the dashboard

1. Open Power BI Desktop and load your healthcare dataset.
2. Create the required DAX measures.
3. Add the four visuals to the report canvas.
4. Resize and arrange them in a two-by-two grid as shown above.
5. Apply consistent colors, titles, and data labels.
6. Add slicers for Department, Region, and Date so users can filter the dashboard.
7. Use Format → Edit interactions to ensure the visuals respond appropriately to selections.

3. Required DAX measures

- Total Treatment Cost =
SUM('Healthcare'[Treatment Cost])
- Average Satisfaction Score =
AVERAGE('Healthcare'[Satisfaction Score])
- Male Patients =
- CALCULATE(
- DISTINCTCOUNT('Healthcare'[Patient ID]),
- 'Healthcare'[Gender] = "Male"
)
- Female Patients =
- CALCULATE(
- DISTINCTCOUNT('Healthcare'[Patient ID]),
- 'Healthcare'[Gender] = "Female"
)

Replace `'Healthcare'` and the column names with those in your dataset if necessary.

4. Visual configuration

Visualization	Fields

Heatmap	Rows: Department; Columns: Region; Values: Total Treatment Cost
Dual Axis Chart	X-axis: Department; Columns: Total Treatment Cost; Line: Average Satisfaction Score
Sunburst Chart	Hierarchy: Department → Diagnosis; Values: Total Treatment Cost
Butterfly Chart	Category: Department; Left: Male Patients; Right: Female Patients

For the Sunburst and Butterfly charts, install a suitable custom visual from Microsoft AppSource if your Power BI version does not provide one natively.

5. Expected insights

- Heatmap: Identifies departments and regions with unusually high treatment costs.
- Dual Axis Chart: Compares treatment expenditure with patient satisfaction.
- Sunburst Chart: Identifies diagnoses contributing most to departmental treatment costs.
- Butterfly Chart: Highlights differences in male and female patient counts across departments.

Conclusion: This dashboard brings together four complementary visualizations to help healthcare managers monitor costs, evaluate patient satisfaction, understand diagnosis-level spending, and compare patient demographics in one interactive report.